PROJECT PROPOSAL
CAMPUS LOST AND FOUND SYSTEM

A Centralized Desktop Database Matching Tool

Prepared by:

Aliha Anwar Rana – 250901030

Minahil Aamir - 250901026

Amna Khurshid – 250901041

Chaudhry Hassan Raza - 250901025

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Submitted to: Miss Fizza

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1. Project Overview

The Campus Lost and Found System is a digital matching desk for a university campus. Think of it as an automated notebook that security guards, students, and administration can use to log misplaced items (like keys, phones, or wallets) and find out if someone has already reported or recovered them. Instead of searching through paper logs or scrolling through social media pages, users interact with a simple desktop menu that does the tracking, storing, and matching automatically.

2. Real-World Problem & Solution

The Problem: When items are lost on a busy campus, information is scattered. The person who lost a phone doesn't know that a finder already dropped it off at a security desk. Additionally, generic spreadsheets often mix up different details, such as cash rewards for lost items versus locker numbers where found items are being held.

The Solution: This application creates two distinct forms—one specifically for losing an item and one for finding an item. It then runs an internal search engine that actively links the two sides together.

3. Blueprint of the Code (The Classes)

To keep the program organized, the code uses a 'parent and child' blueprint system:

- **The Parent Template (Item):** This is a general blueprint that holds basic information common to every object, such as the item's name, where it was spotted, and a contact phone number. You cannot create a blank 'item' on its own; it only exists to pass these traits down.
- **Child Template 1 (LostItem):** A specialized form used when a student loses something. It inherits all the basic traits from the parent but adds a specific line for tracking a monetary reward if the owner wants to offer one.
- **Child Template 2 (FoundItem):** A specialized form used when someone turns an item in. It inherits the basic traits but adds a specific line for tracking a physical storage bin number so the campus administration knows exactly which locker it is being stored in.

4. How the Computer Remembers Data (Memory Types)

Your computer splits its memory into two different zones to manage this application:

Memory Type	What It Is	How It Is Used In This Project
The Stack	A temporary notepad the computer uses to take quick notes or handle short tasks.	When a user is actively typing a name or choosing a option, that text sits on the Stack. The moment the task ends, the computer completely wipes that data page away.

Memory Type	What It Is	How It Is Used In This Project
The Heap	A permanent storage locker or filing cabinet.	Because the notepad (Stack) gets wiped constantly, we use the 'new' keyword to save final item records into the Heap. Data here stays locked away safely for the entire duration of the program.

5. Summary of How it Works (Core Features)

- **The Dashboard:** A simple 1-to-5 numeric menu lets users select whether they want to report a loss, log a recovery, view the database, or run a search.
- **Smart Filtering:** If a user accidentally types letters (like 'abc') into the number menu, the program catches the typo, flushes the bad text away, and smoothly reloads the menu instead of crashing.
- **The Matchmaker:** When you run the cross-reference feature, the program scans the filing cabinet. It ignores matches between two lost items or two found items. It only alerts the user if a Lost Item name perfectly matches a Found Item name, instantly printing the owner's and finder's contact details side-by-side on screen.